

Original Issue
Revision :

19 August, 2011
N/R



STARTER GENERATOR INSTALLATION MANUAL

STARTER GENERATOR SERIES:

150SG	250SG
160SG	300SG
200SG	300SGL
200SGL	400SG

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RECORD OF REVISIONS

Revision Number	Date Issued	Date Inserted Into Manual	Initials
Original Issue	19 August, 2011		Factory
Retain this record in front of Manual. Only the latest revision level of this manual is considered valid for current work being performed on SAI starter generators. Operators/maintainers shall dispose of superseded material in accordance with their own technical documentation procedures.			



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SERVICE BULLETIN CONTROL

Service Bulletin Number	Revision Level	Latest Rev. Date	Status
SB 150SG114	C	Dec. 9, 2009	Incorporated
Retain this record in front of Manual. On receipt of Service Bulletins, record service bulletin number, status (valid, invalid, etc.), revision, and date issued. Unless invalidated, all service bulletins affecting SAI starter generators should be kept with this manual.			



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SERVICE LETTER CONTROL

Service Letter Number	Revision Level	Latest Rev. Date	Status
134	B	Dec. 9, 2009	Incorporated
115	B	Dec. 9, 2009	Incorporated
225	N/R	Jan. 25, 2011	Incorporated
Retain this record in front of Manual. On receipt of Service Letters, record the Service Letter number, status (valid, invalid, etc.), revision, and date issued. Unless invalidated, all Service Letters affecting the SAI starter generators should be kept with this Manual.			



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TECHNICAL DOCUMENTATION SUGGESTION FORM



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DESCRIPTION OF REVISIONS

Page No.	Description	Revision Date	Approved
Entire Manual	Initial issue.	5 August, 2011	



AL-1 AIRWORTHINESS LIMITATIONS

The Airworthiness Limitations section is FAA compliant and specifies maintenance required under 14 CFR §§ 43.16 and 91.403, unless an alternative program has been FAA approved.

There are no airworthiness limitations for this series of starter generators.

AL-2 RECOMMENDED OVERHAUL PERIOD

Unless established otherwise in the aircraft operator's own Airworthiness Limitations or Operating Specifications, the recommended overhaul period for SAI starter-generators is 1,000 hours with a minimum of 1 brush change prior.

AL-3 FIELD SERVICING LIMITS

Recommended field service on starter-generator units is limited to brush inspection only. All other repair and/or overhaul activity should be carried out by an authorized repair station.

Except for brush inspections, aircraft airworthiness is accomplished through removal and replacement of the starter-generator.

NOTE

Consult the Service Letters section of this manual for information on brush life calculations.

AL-4 START CYCLES

After every start cycle, the starter-generator as well as the engine should have the opportunity to cool off. The minimum cooling times are stated below:

30 seconds ON – 1 minute OFF
30 seconds ON – 1 minute OFF
30 seconds ON – 30 minutes OFF

AL-5 STORAGE LIMITS

If the starter-generator has been in storage for longer than 24 months, it should not be installed on an aircraft. It must be returned to an SAI authorized repair facility for testing and if necessary, repair.

Bearing grease shelf life is 5 years from date of manufacture of the bearing.

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SECTION I

INTRODUCTION

1-1 GENERAL

- 1-2 This manual contains introductory information and installation data for Skurka Aerospace Inc. starter-generators. These starter-generators are manufactured by:

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NOTE:

If an error, omission, or other discrepancy is found in this manual, please fill out a copy of the Technical Documentation Suggestion Form found at the back of this manual and forward to the attention of "Sales—Product Line Manager" at Skurka Aerospace Inc.

1-2a CHANGES TO STARTER-GENERATOR

- 1-2b Changes to the starter-generator that may affect this manual are communicated and distributed by manual revisions, and by Service Bulletins (SB's) or Service Letters (SL's). SB's and SL's may be incorporated at the next revision change when applicable. Active or valid service information pertinent to this manual is located in the appropriate section of the manual. Any service information issued after the effective date of this manual should be documented on the appropriate control sheet in the Table of Contents and kept with this manual.

1-3 PURPOSE OF EQUIPMENT

- 1-4 The Starter-Generator is used as an engine-driven source of DC power in a 28 volt aircraft electrical system and as a source of starting power for light-weight turbine engines. An efficient integral fan draws air in at the anti-drive end of generator to permit self-cooled operation.

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SECTION II

INSTALLATION INSTRUCTIONS

Expanded procedures have been developed to remove or install the starter-generator in conjunction with the procedures provided in current applicable maintenance manuals. These procedures should be utilized each time the starter-generator is removed or installed to assist in improving the service life of the starter generator bearings, brushes and drive shaft friction disc.

NOTE:

In all cases, refer to the applicable aircraft maintenance manual for proper torque, safetying and other detailed removal and installation instructions that may be particular to the aircraft model. The information in the aircraft maintenance manual should be used if there is a conflict with information contained in these instructions.

Two mechanics are required to properly remove or install the starter-generator.

CAUTION

Slight misalignment and/or binding of the starter-generator driveshaft can reduce the unit's service life.

1. Removal of Starter-Generator

- a. Disconnect power and then electrical connections per aircraft manual.
- b. One mechanic is to maintain the alignment of the starter-generator by holding the unit in position by keeping the starter generator mounting surfaces flush with the Quick Attach/Detach adapter (QAD pad), while the other mechanic loosens and removes the 'V' band clamp.
- c. Carefully remove the starter-generator from the QAD pad by maintaining the spline's alignment to prevent binding of the starter-generator driveshaft.



2. Prior to Installation, Verify:

- a. There are no burrs or foreign objects on the starter-generator shaft.
- b. The starter-generator guide pins are clean and not bent or damaged.
- c. The mounting surfaces of the starter-generator and the QAD pad are clean and free of burrs.
- d. The QAD pad is properly secured to the engine.
- e. The QAD pad guide pin holes are free of burrs, foreign objects and in good condition.
- f. For wet spline starter-generators check 'O' ring on drive shaft to see if it is worn or cut. Replace if necessary.
- g. Drive shaft is concentric to the armature of the starter-generator (See Fig. 1). If the drive shaft appears to be out of position, lightly tap on the drive shaft spline, using a soft faced mallet to move it back into full concentric position.

CAUTION

Do not use driveshaft spline as a grip to lift starter generator.

- h. Verify V band clamp nut tightening torque from band clamp surface.

3. Installation of the Starter-Generator

- a. One mechanic is to gently align the starter-generator to the QAD pad, being careful not to put the starter-generator drive spline in a bind.
- b. After the starter-generator has been properly aligned, one mechanic should hold the unit in position by keeping the starter generator mounting surfaces flush with the QAD pad while the other mechanic installs and secures the 'V' band clamp (note the installation torque on clamp ring).

While tightening the nut on the T-bolt to its recommended torque value, tap around the circumference of the clamp ring with a soft faced hammer to position the clamp firmly on the generator.

The recommended torque value for the clamp is stamped on the clamp ring.



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4. Electrical Connections

- a. Make all terminal block electrical connections between the aircraft wiring and the starter-generator, as indicated in the applicable aircraft maintenance manual. Recommended torque values for terminal block connections are shown in Table I.

TABLE I

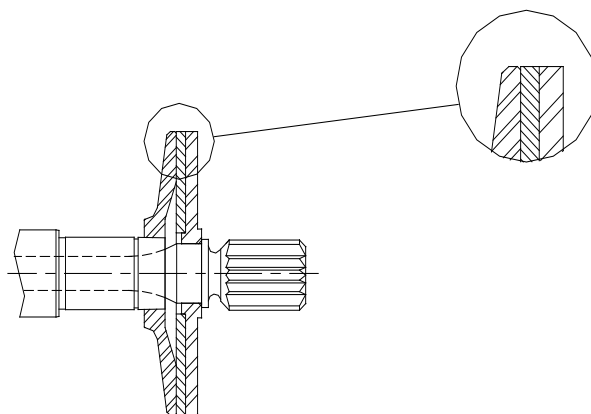
**RECOMMENDED TERMINAL BLOCK ELECTRICAL
CONNECTION NUT TORQUE VALUES**

Terminal Size	Torque Value (pound-inches)
# 3/8-24 Stud	160 – 190*
# 10-32 Stud	20 – 25

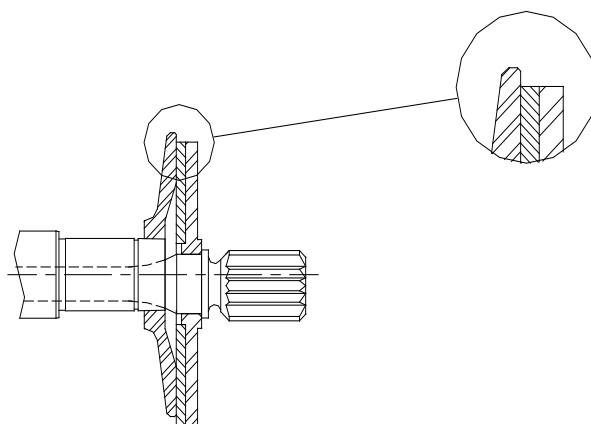
*May be increased to 200-235 in lbs. if needed for multiple wires on a single terminal.



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DRIVE SHAFT CONCENTRIC WITH ARMATURE SHAFT



DRIVE SHAFT OUT OF POSITION

Figure 1.

Drive Shaft Alignment



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SECTION III

SERVICE BULLETINS

No installation related Service Bulletins apply as of this manual's date of issuance.

Applicable Service Bulletins issued after this manual's issuance date should be kept in this section and recorded on the Service Bulletin Control page.

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SECTION IV

SERVICE LETTERS

SL 115	Brush Life Calculations for 150/160SG
SL 132	Brush Life Calculations for 200SGL
SL 178	Brush Life Calculations for 200/250SG
SL 193	Brush Life Calculations for 300SG/300SGL
SL 214	Brush Life Calculations for 400SG



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SERVICE LETTER NO.: SL115

**TO: ALL HOLDERS OF SKURKA AEROSPACE INC. (SAI) 150SG and 160SG
SERIES STARTER-GENERATORS**

FROM: S. SHERMAN

**SUBJECT: DATA WHICH WILL ALLOW YOU TO CALCULATE YOUR BRUSH LIFE
FOR YOUR SAI STARTER-GENERATORS**

We have found that repeated brush inspections of Starter-Generators can result in rapid brush wear and even hang-ups. When inspecting brushes, many operators will slide the brush cover off the back of the generator causing the brush leads to be misplaced.

Sliding the cover forward again, further misplaces the brush leads. If the generator is mounted on the engine while the inspection is being made, some brush boxes cannot be seen, making brush lead positioning impossible.

Removing brushes from brush boxes can cause the brush position in the box and the relationship to the commutator of the brush to be disturbed. The disturbed brush may wear faster than normal.

One brush inspection between overhauls of the SAI generator is all that may be required of the SAI brushes.

In essence, this procedure allows the operator to determine his time between brush changes and may help fix the overhaul periods to be at the same time brushes must be replaced.

At 300 hours, remove the generator for brush inspection. Remove the shortest brush only to measure the remaining brush life. Do not remove more than one brush, as brush seating will be affected. Your brush inspection time will be obtained by measuring three different generators and using the shortest brush length, as the basis for determining your fleet wide brush life. Filling in the attached form, will show you how to determine remaining brush life. Using this procedure will aid you in determining your Mean Time Between Overhauls (MTBO). The generator must be overhauled at 1,000 hours, at which time the bearing must be replaced.

The SAI 150SG and 160SG ampere Starter-Generators can usually operate for a minimum of 500-800 hours without brush change. At 1,000 hours time, the generator must be removed for overhaul and bearings changed, unless the SAI overhaul manual or aircraft manufacturer states the bearings can be operated in excess of 1,000 hours.

Please note, this procedure can only be accomplished with a new generator or a generator that has been overhauled in accordance with SAI Overhaul Procedures and the commutator has been turned (resurfaced) in a lathe.



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GENERATOR SERIES 150SG/160SG SAMPLE CALCULATION PROCEDURE

At 300 hours, remove the generator for brush inspection. Remove the shortest brush only. Do not remove more than one brush as brush seating will be affected.

SAMPLE CALCULATION

Date: 01/09/97

Generator P/N: 150SG111Q

Serial No.: 1200

Calculate brush life left as per example below:

Maximum wear length of a new 150SG1009-5 Brush (1) .520

Brush wear left (as measured from shortest brush) (2) .300

TOTAL BRUSH WEAR (3) .220

Hours brush has operated: (4) 300

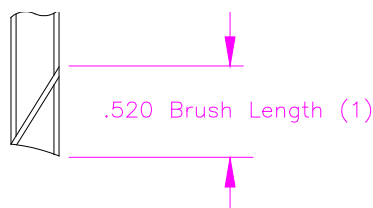
(3) $\frac{.220}{\text{Total Brush Wear (from \#3 above)}}$ $\div$ (4) $\frac{300}{\text{Hours Brush Operated (from \#4 above)}}$ = (5) .0007 Rate of brush wear per hour

(2) $\frac{.300}{\text{Brush Wear Left (from Item \#2 above)}}$ $\div$ (5) $\frac{.0007}{\text{Rate of Brush Wear per Hour (from Item \#5 above)}}$ = (6) 429 Hours of brush left before brushes must be replaced

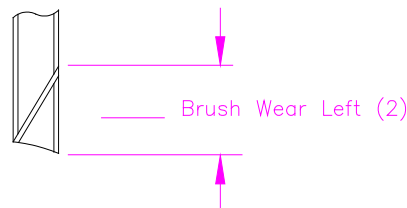
Generator Operating Hours: $\frac{429}{\text{Hours Left (from Item \#6)}}$ + $\frac{300}{\text{Total Hours on Generator Thus Far (from Item \#4)}}$ = $\frac{729}{\text{Total Hours Generator Can Operate Before Brush Change}}$

If 150 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above or for a maximum of 1,000 hours. At that time, return the unit to an Authorized SAI Repair Station for overhaul.

New Brush



Shortest Brush from Generator



Remove the generator the second time after the additional operating hours established by your calculations has been reached. Inspect the brushes. If all are below the maximum brush wear mark, the generator MTBO has been established. If you desire, two or three more generators can be tested to establish an average brush life.



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GENERATOR SERIES 150SG/160SG SAMPLE CALCULATION PROCEDURE

At 300 hours, remove the generator for brush inspection. Remove the shortest brush only. Do not remove more than one brush as brush seating will be affected.

Date: _____

Generator P/N: _____

Serial No.: _____

Calculate brush life left as per example below:

Maximum wear length of a new 150SG1009-5 Brush (1) .520

Brush wear left (as measured from shortest brush) (2) _____

TOTAL BRUSH WEAR (3) _____

Hours brush has operated: (4) _____

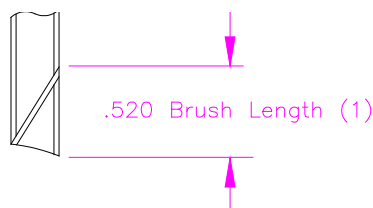
(3) $\frac{\text{Total Brush Wear (from \#3 above)}}{\text{Hours Brush Operated (from \#4 above)}} =$ (5) _____ Rate of brush wear per hour

(2) $\frac{\text{Brush Wear Left (from Item \#2 above)}}{\text{Rate of Brush Wear per Hour (from Item \#5 above)}} =$ (6) _____ Hours of brush left before brushes must be replaced

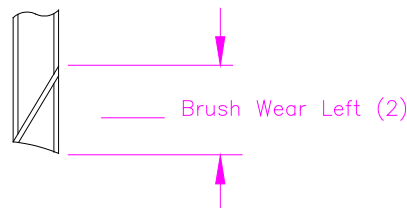
Generator Operating Hours: _____ + $\frac{\text{Hours Left (from Item \#6)}}{\text{Total Hours on Generator Thus Far (from Item \#4)}} =$ $\frac{\text{Total Hours Generator Can Operate Before Brush Change}}$

If 150 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above or for a maximum of 1,000 hours. At that time, return the unit to an Authorized APC Repair Station for overhaul.

New Brush



Shortest Brush from Generator



Remove the generator the second time after the additional operating hours established by your calculations has been reached. Inspect the brushes. If all are below the maximum brush wear mark, the generator MTBO has been established. If you desire, two or three more generators can be tested to establish an average brush life.



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SERVICE LETTER NO.: SL132

TO: ALL HOLDERS OF SAI 200SGL SERIES STARTER-GENERATORS

**SUBJECT: DATA WHICH WILL ALLOW YOU TO CALCULATE YOUR BRUSH LIFE
FOR YOUR SAI STARTER-GENERATORS**

We have found that repeated brush inspections of Starter-Generators can result in rapid brush wear and even hang-ups. When inspecting brushes, many operators will slide the brush cover off the back of the generator causing the brush leads to be misplaced.

Sliding the cover forward again, further misplaces the brush leads. If the generator is mounted on the engine while the inspection is being made, some brush boxes cannot be seen, making brush lead positioning impossible.

Removing brushes from brush boxes can cause the brush position in the box and the relationship to the commutator of the brush to be disturbed. The disturbed brush may wear faster than normal.

One brush inspection between overhauls of the SAI generator is all that may be required of the SAI brushes.

In essence, this procedure allows the operator to determine his time between brush changes and may help fix the overhaul periods to be at the same time brushes must be replaced.

At 300 hours, remove the generator for brush inspection. Remove the shortest brush only to measure the remaining brush life. Do not remove more than one brush, as brush seating will be affected. Your brush inspection time will be obtained by measuring three different generators and using the shortest brush length, as the basis for determining your fleet wide brush life. Filling in the attached form, will show you how to determine remaining brush life. Using this procedure will aid you in determining your Mean Time Between Overhauls (MTBO). The generator must be overhauled at 1,000 hours, at which time the bearing must be replaced.

The SAI 200SGL ampere Starter-Generators can usually operate for a minimum of 400-800 hours without brush change. At 1,000 hours time, the generator must be removed for overhaul and bearings changed, unless the SAI overhaul manual or aircraft manufacturer states the bearings can be operated in excess of 1,000 hours.

Please note, this procedure can only be accomplished with a new generator or a generator that has been overhauled in accordance with SAI Overhaul Procedures and the commutator has been turned (resurfaced) in a lathe.



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GENERATOR SERIES 200SGL SAMPLE CALCULATION PROCEDURE BRUSH P/N 200SGL1009

At 300 hours, remove the generator for brush inspection. Remove the shortest brush only. Do not remove more than one brush as brush seating will be affected.

SAMPLE CALCULATION

Date: 16 Dec 1998

Generator P/N: 200SGL117Q

Serial No.: 1200

Calculate brush life left as per example below:

Maximum wear length of a new 150SG1009-5 Brush (1) .625

Brush wear left (as measured from shortest brush) (2) .310

TOTAL BRUSH WEAR (3) .315

Hours brush has operated: (4) 300

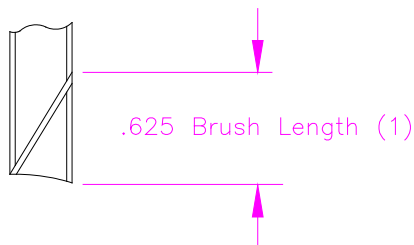
(3) $\frac{.315}{\text{Total Brush Wear (from \#3 above)}}$ $\div$ (4) $\frac{300}{\text{Hours Brush Operated (from \#4 above)}}$ = (5) .00105 Rate of brush wear per hour

(2) $\frac{.310}{\text{Brush Wear Left (from Item \#2 above)}}$ $\div$ (5) $\frac{.00105}{\text{Rate of Brush Wear per Hour (from Item \#5 above)}}$ = (6) 295 Hours of brush left before brushes must be replaced

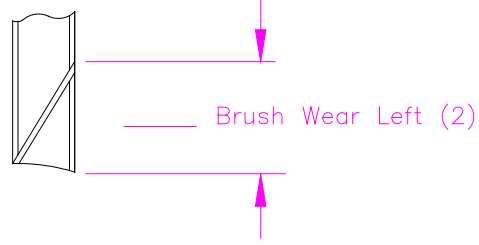
Generator Operating Hours: $\frac{295}{\text{Hours Left (from Item \#6)}}$ + $\frac{300}{\text{Total Hours on Generator Thus Far (from Item \#4)}}$ = $\frac{595}{\text{Total Hours Generator Can Operate Before Brush Change}}$

If 150 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above or for a maximum of 1,000 hours. At that time, return the unit to an Authorized SAI Repair Station for overhaul.

New Brush



Shortest Brush from Generator



Remove the generator the second time after the additional operating hours established by your calculations has been reached. Inspect the brushes. If all are below the maximum brush wear mark, the generator MTBO has been established. If you desire, two or three more generators can be tested to establish an average brush life.



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GENERATOR SERIES 200SGL SAMPLE CALCULATION PROCEDURE BRUSH P/N 200SGL1009

At 300 hours, remove the generator for brush inspection. Remove the shortest brush only. Do not remove more than one brush as brush seating will be affected.

Date: _____

Generator P/N: _____

Serial No.: _____

Calculate brush life left as per example below:

Maximum wear length of a new 150SG1009-5 Brush (1) .625

Brush wear left (as measured from shortest brush) (2) _____

TOTAL BRUSH WEAR (3) _____

Hours brush has operated: (4) _____

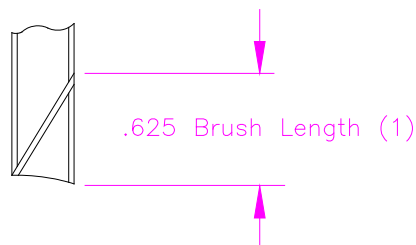
(3) $\frac{\text{Total Brush Wear (from \#3 above)}}{\text{Hours Brush Operated (from \#4 above)}} =$ (5) _____ Rate of brush wear per hour

(2) $\frac{\text{Brush Wear Left (from Item \#2 above)}}{\text{Rate of Brush Wear per Hour (from Item \#5 above)}} =$ (6) _____ Hours of brush left before brushes must be replaced

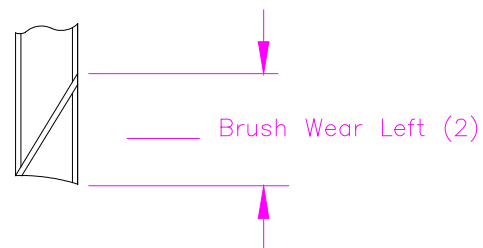
Generator Operating Hours: _____ + _____ = _____
Hours Left (from Item #6) Total Hours on Generator Thus Far (from Item #4) Total Hours Generator Can Operate Before Brush Change

If 150 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above or for a maximum of 1,000 hours. At that time, return the unit to an Authorized SAI Repair Station for overhaul.

New Brush



Shortest Brush from Generator



Remove the generator the second time after the additional operating hours established by your calculations has been reached. Inspect the brushes. If all are below the maximum brush wear mark, the generator MTBO has been established. If you desire, two or three more generators can be tested to establish an average brush life.



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SERVICE LETTER NO.: SL178

TO: ALL USERS OF SAI NEW 200SG & 250SG SERIES STARTER-GENERATORS

SUBJECT: DATA WHICH WILL ALLOW YOU TO CALCULATE YOUR BRUSH LIFE FOR YOUR SAI STARTER-GENERATOR

We have found that repeated brush inspections of starter-generators, can result in rapid brush wear and even cause brush hang-ups. When inspecting brushes, many operators will slide the brush cover off the back of the generator causing the brush leads to be misplaced.

Sliding the cover forward again further misplaces the brush leads. If the generator is mounted on the engine while the inspection is being made, some brush boxes cannot be seen, making brush lead positioning impossible.

Removing brushes from brush boxes will cause brush position in the box and relationship to the commutator to be disturbed. The mispositioned brush will wear faster than normal.

One brush inspection between overhauls of the SAI generator is all that may be required of SAI brushes.

That one inspection can possibly be eliminated if the accompanying procedure is adhered to.

In essence, this procedure allows you to determine the time between brush changes and helps fix overhaul periods at the same time brushes must be replaced.

The SAI 200 or 250 ampere Starter-Generator can usually operate for 800 – 1000 hours without brush change. At that time, the generator can be removed for overhaul. Using this procedure will aid you in determining your overhaul times.

Please note, this procedure can only be accomplished with a new generator or a generator that has been overhauled in accord with SAI Overhaul Procedures and the commutator has been turned (resurfaced) in on lathe.



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GENERATOR SERIES 200SG & 250SG *SAMPLE CALCULATION PROCEDURE* BRUSH P/N 200SG1009 & 250SG1009

At 500 hours, remove the generator for brush inspection. Remove the shortest brush only. Do not remove more than one brush as brush seating will be affected.

Date: 19 FEB 1990

Generator P/N: 250SG117Q

Serial No.: 1200

Calculate brush life left as per example below:

Maximum wear length of a new 200SG or 250SG1009 Brush (1) .700

Brush wear left (as measured from shortest brush) (2) .375

TOTAL BRUSH WEAR (3) .325

Hours brush has operated: (4) 500

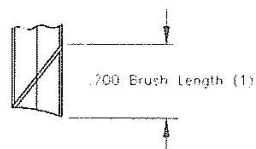
(3) .325 ÷ (4) 500 = (5) .00065 Rate of brush wear per hour
Total Brush Wear (from #3 above) Hours Brush Operated (from #4 above)

(2) .375 ÷ (5) .00065 = (6) 577 Hours of brush left before brushes must be replaced
Brush Wear Left (from Item #2 above) Rate of Brush Wear per Hour (from Item #5 above)

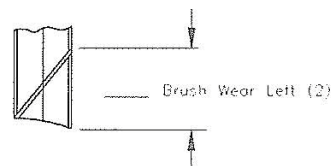
Generator Operating Hours: 577 + 500 = 1077
Hours Left (from Item #6) Total Hours on Generator Thus Far (from Item #4) Total Hours Generator Can Operate Before Brush Change

If 150 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above or for a maximum of 1,000 hours. At that time, return the unit to an Authorized SAI Repair Station for overhaul.

NEW BRUSH



SHORTEST BRUSH FROM GENERATOR



Remove the generator the second time after the additional operating hours established by your calculations has been reached. Inspect the brushes. If all brushes are below the maximum brush wear mark, the generator MTBO has been established. If you desire, two or three generators can be tested to establish an average brush life.



Skurka Aerospace Inc.
4600 Calle Bolero, P.O. Box 2869
Camarillo, CA 93011

GENERATOR SERIES 200SG & 250SG, BRUSH P/N 200SG1009 & 250SG1009

At 500 hours, remove the generator for brush inspection. Remove the shortest brush only. Do not remove more than one brush as brush seating will be affected.

Date: _____

Generator P/N: _____

Serial No.: _____

Calculate brush life left as per example below:

Maximum wear length of a new 200SG or 250SG1009 Brush (1) _____

Brush wear left (as measured from shortest brush) (2) _____

TOTAL BRUSH WEAR (3) _____

Hours brush has operated: (4) _____

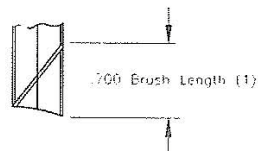
(3) _____ ÷ (4) _____ = (5) _____ Rate of brush wear per hour
Total Brush Wear (from #3 above) Hours Brush Operated (from #4 above)

(2) _____ ÷ (5) _____ = (6) _____ Hours of brush left before brushes must be replaced
Brush Wear Left (from Item #2 above) Rate of Brush Wear per Hour (from Item #5 above)

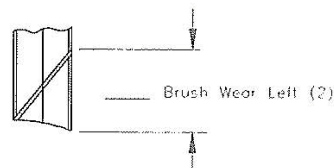
Generator Operating Hours: _____ + _____ = _____
Hours Left (from Item #6) Total Hours on Generator Thus Far (from Item #4) Total Hours Generator Can Operate Before Brush Change

If 150 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above or for a maximum of 1,000 hours. At that time, return the unit to an Authorized **SAI** Repair Station for overhaul.

NEW BRUSH



SHORTEST BRUSH FROM GENERATOR



Remove the generator the second time after the additional operating hours established by your calculations has been reached. Inspect the brushes. If all are below the maximum brush wear mark, the generator MTBO has been established. If you desire, two or three generators can be tested to establish an average brush life.



Skurka Aerospace Inc.
4600 Calle Bolero; P.O Box 2869
Camarillo, CA 93011

SERVICE LETTER NO.: SL193

ATTENTION: ALL HOLDERS OF NEW 300SG & 300SGL SERIES STARTER-GENERATORS

**SUBJECT: BRUSH LIFE CALCULATIONS AT FIRST INSPECTION OF THE
STARTER-GENERATOR**

This inspection allows you to determine the time between brush changes and helps fix overhaul periods at the same time brushes must be replaced.

The SAI 300 ampere Starter-Generator can usually operate for 800 - 1000 hours without brush change. At that time, the generator can be removed for overhaul. Using this procedure will aid you in determining your overhaul times.

Please note, this procedure can only be accomplished with a new generator or a generator that has been overhauled in accord with SAI Overhaul Procedures and the commutator has been turned (resurfaced) in on lathe.



ACCOMPLISHMENT INSTRUCTIONS

CAUTION: USE CARE WHEN MOVING THE COVER FORWARD AS IT WILL MISPLACE THE BRUSH LEADS. WHEN THE STARTER GENERATOR IS MOUNTED ON THE ENGINE WHILE THE INSPECTION IS BEING MADE, SOME BRUSH LEADS CANNOT BE SEEN.

1. Remove Brush Strap. See Figure 1. (Note position of screw head.)



Figure 1
Brush Strap

2. Remove Brush screw from Brush Set #1. (Terminal Block to left of brush set.) See Figure 2.
3. Gently release Brush Springs and remove Brush Set #1.

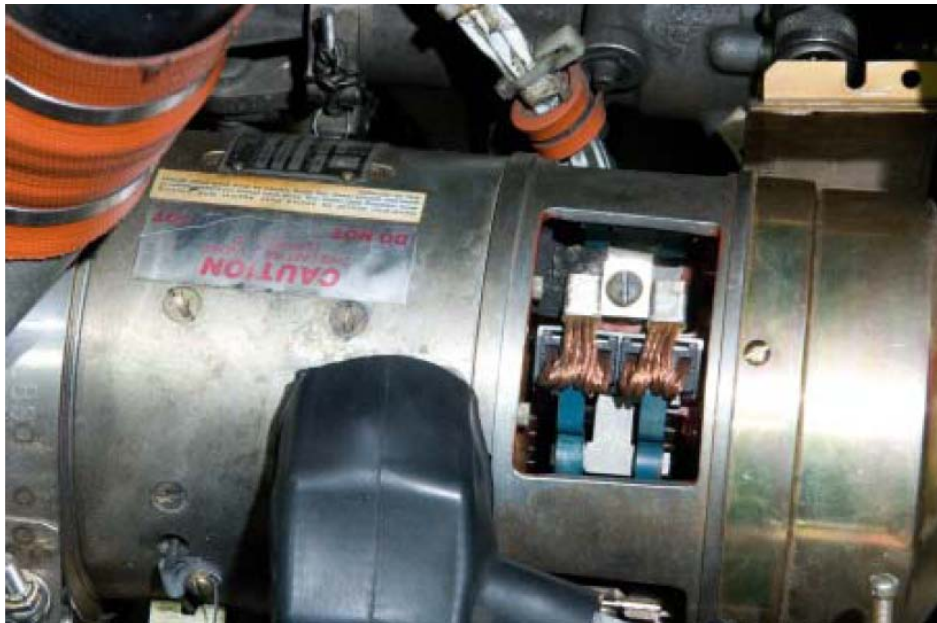


Figure 2
Brush Set #1

4. Use a flashlight to visibly inspect the condition of the commutator surface. If severe damage (i.e., arced, pitted, etc.) is noted, then replace unit. See Figure 3 for an example of a damaged commutator.

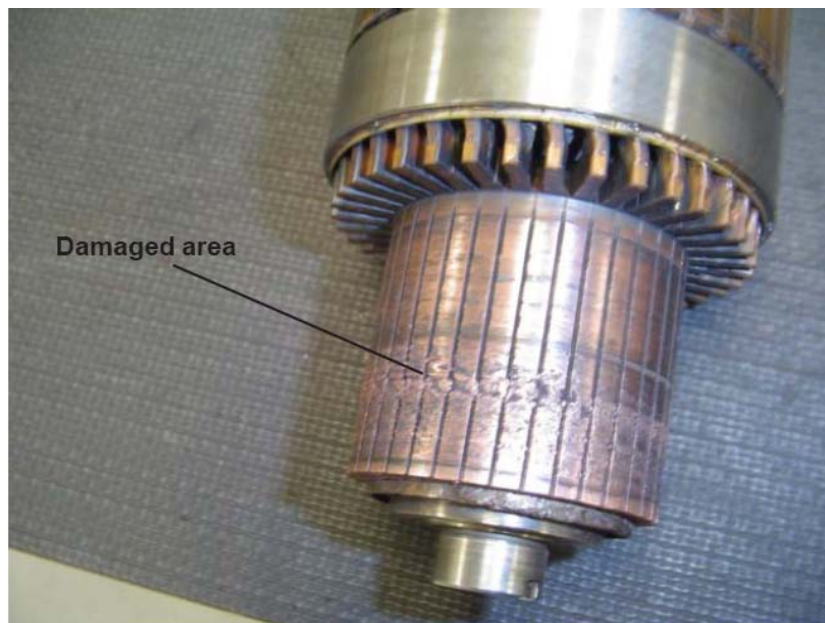


Figure 3
Damaged Commutator



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5. Using calipers, measure from the wear indicator line to the end of the brush following the figures on page 5 of the bulletin as guide.
6. Use pages 5 and 6 of this Service Letter to calculate remaining hours of operation.

CAUTION: MAKE SURE NOT TO SNAP THE BRUSH SPRINGS. THIS CAN CAUSE DAMAGE TO THE BRUSH SET AND DECREASE BRUSH LIFE.

7. If there are sufficient hours are left, reinstall Brush Set #1. Refer to Figure 4 of this Service Letter for proper brush lead position. If there are not sufficient hours left, then replace unit.

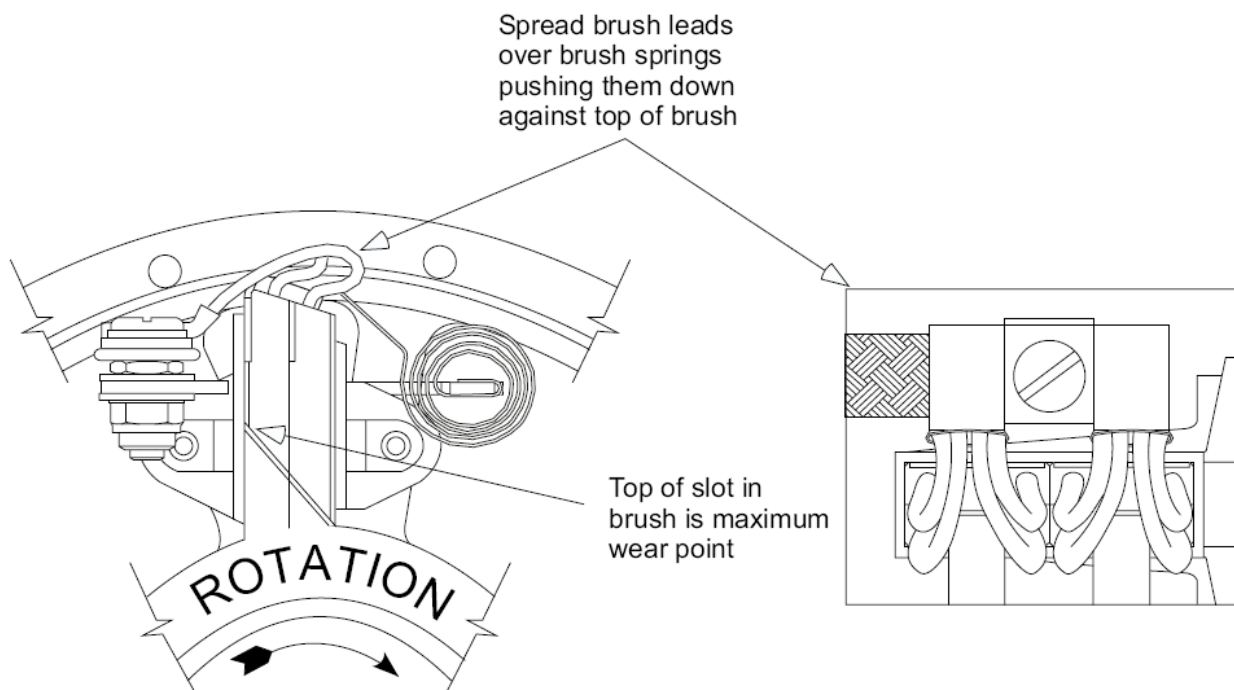


Figure 4
Positioning of Brush Leads

8. Finally, reinstall Brush Strap. (Note position of screw head.)



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Camarillo, CA 93011

GENERATOR SERIES 300SG, BRUSH P/N 300SGL1009-5
SAMPLE CALCULATION PROCEDURE

Date: _____

Generator P/N: _____

Serial No.: _____

Calculate brush life left as per example below:

Maximum wear length of a new 300SGL1009-5 Brush (1) .700

Brush wear left (as measured from brush set #1) (2) .375

TOTAL BRUSH WEAR (3) .325

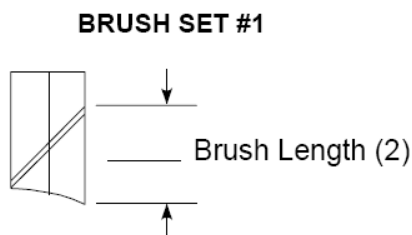
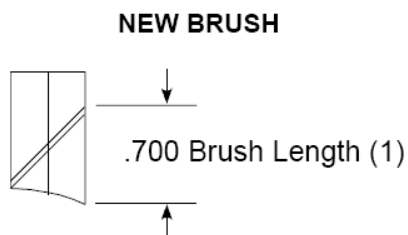
Hours brush has operated: (4) 500

(3) .325 ÷ (4) 500 = (5) .00065 Rate of brush wear per hour
Total Brush Wear (from #3 above) Hours Brush Operated (from #4 above)

(2) .375 ÷ (5) .00065 = (6) 577 Hours of brush left before brushes must be replaced
Brush Wear Left (from Item #2 above) Rate of Brush Wear per Hour (from Item #5 above)

Generator Operating Hours: 577 + 500 = 1077
Hours Left (from Item #6) Total Hours on Generator Thus Far (from Item #4) Total Hours Generator Can Operate Before Brush Change

If 250 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above. At that time, return the unit to Skurka Aerospace Inc. for overhaul.





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Camarillo, CA 93011

GENERATOR SERIES 300SG AND 300SGL

Date: _____

Generator P/N: _____

Serial No.: _____

Calculate brush life left as per example below:

Maximum wear length of a new 300SGL1009-5 Brush (1) .700

Brush wear left (as measured from brush set # 1) (2) _____

TOTAL BRUSH WEAR (3) _____

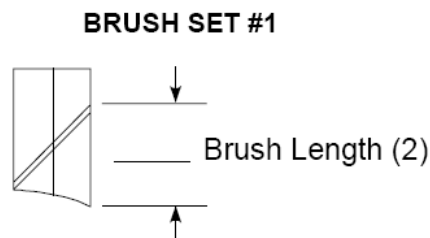
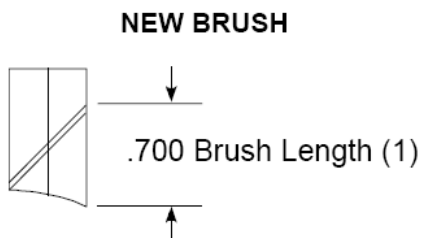
Hours brush has operated: (4) _____

(3) _____ ÷ (4) _____ = (5) _____ Rate of brush wear per hour
Total Brush Wear (from #3 above) Hours Brush Operated (from #4 above)

(2) _____ ÷ (5) _____ = (6) _____ Hours of brush left before brushes must be replaced
Brush Wear Left (from Item #2 above) Rate of Brush Wear per Hour (from Item #5 above)

Generator Operating Hours: _____ + _____ = _____
Hours Left (from Item #6) Total Hours on Generator Thus Far (from Item #4) Total Hours Generator Can Operate Before Brush Change

If 250 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above. At that time, return the unit to Skurka Aerospace Inc. for overhaul.





SERVICE LETTER # 214

DATA WHICH WILL ALLOW YOU TO CALCULATE YOUR BRUSH LIFE FOR YOUR SAI STARTER-GENERATOR

We have found that repeated brush inspections of starter-generators, can result in rapid brush wear and even cause brush hang-ups. When inspecting brushes, many operators will slide the brush cover off the back of the generator causing the brush leads to be misplaced.

Sliding the cover forward again further misplaces the brush leads. If the generator is mounted on the engine while the inspection is being made, some brush boxes cannot be seen, making brush lead positioning impossible.

Removing brushes from brush boxes will cause brush position in the box and relationship to the commutator to be disturbed. The mispositioned brush will wear faster than normal.

One brush inspection between overhauls of the SAI generator is all that may be required of SAI brushes.

That one inspection can possibly be eliminated if the accompanying procedure is adhered to.

In essence, this procedure allows you to determine the time between brush changes and helps fix overhaul periods at the same time brushes must be replaced.

The SAI 400 ampere Starter-Generator can usually operate for 800 - 1000 hours without brush change. At that time, the generator can be removed for overhaul. Using this procedure will aid you in determining your overhaul times.

Please note, this procedure can only be accomplished with a new generator or a generator that has been overhauled in accord with SAI Overhaul Procedures and the commutator has been turned (resurfaced) in on lathe.



Skurka Aerospace Inc.
4600 Calle Bolero, P. O. Box 2869
Camarillo, CA 93011

GENERATOR SERIES 400SG SAMPLE CALCULATION PROCEDURE

(Brush P/N 400SG1009-5 used as example)

At 500 hours, remove the generator for brush inspection. Remove the shortest brush only. Do not remove more than one brush as brush seating will be affected.

Date: _____

Generator P/N: _____

Serial No.: _____

Calculate brush life left as per example below:

Maximum wear length of a new 400SG1009-5 Brush (1) .700

Brush wear left (as measured from shortest brush) (2) .375

TOTAL BRUSH WEAR (3) .325

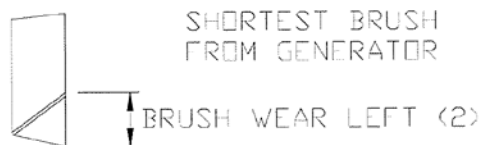
Hours brush has operated: (4) 500

(3) .325 ~ (4) 500 = (5) .00065 Rate of brush wear per hour
Total Brush Hours Brush
Wear (from Operated (from
#2 above) #4 above)

(2) .375 ~ (5) .00065 = (6) 577 Hours of brush wear left
Brush Wear Rate of Brush before brushes must be
Left (from Item Wear per Hour replaced.
#2 above) (from Item #5 above)

Generator Operating Hours: 577 + 500 = 1077
Hours Left Total Hours on Total Hours Generator
from Generator Thus Can Operate Before
Item #6 Far from Item #4 Brush Change

If 150 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above or for a maximum of 1,000 hours. At that time, return the unit to an Authorized SAI Repair Station for overhaul.



Remove the generator the second time after the additional operating hours established by your calculations has been reached. Inspect the brushes. If all brushes are below the maximum brush wear mark, the generator MTBO has been established. If you desire, two or three generators can be tested to establish an average brush life.



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Camarillo, CA 93011

GENERATOR SERIES 400SG CALCULATION PROCEDURE WORKSHEET

At 500 hours, remove the generator for brush inspection. Remove the shortest brush only. Do not remove more than one brush as brush seating will be affected.

Date: _____

Generator P/N: _____

Serial No.: _____

Calculate brush life left as per example below:

Maximum wear length of a new brush. (1) _____

Brush wear left (as measured from shortest brush) (2) _____

TOTAL BRUSH WEAR (3) _____

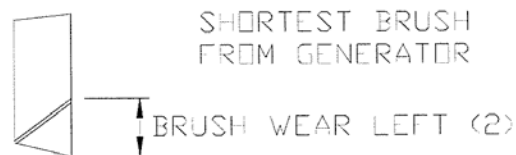
Hours brush has operated: (4) _____

(3) _____ ~ (4) _____ = (5) _____ Rate of brush wear per hour
Total Brush Wear (from #2 above) Hours Brush Operated (from #4 above)

(2) _____ ~ (5) _____ = (6) _____ Hours of brush wear left before brushes must be replaced.
Brush Wear Left (from Item #2 above) Rate of Brush Wear per Hour (from Item #5 above)

Generator Operating Hours: _____ + _____ = _____
Hours Left from Item #6 Total Hours on Generator Thus Far from Item #4 Total Hours Generator Can Operate Before Brush Change

If 150 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above or for a maximum of 1,000 hours. At that time, return the unit to an Authorized SAI Repair Station for overhaul.



Remove the generator the second time after the additional operating hours established by your calculations has been reached. Inspect the brushes. If all are below the maximum brush wear mark, the generator MTBO has been established. If you desire, two or three generators can be tested to establish an average brush life.

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